



ÉCOLE SUPÉRIEURE D'INGÉNIEURS LÉONARD DE VINCI

NURBS SPLATTING

Research Track Report

Bonifas Olivier
olivier.bonifas@edu.devinci.fr

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ABSTRACT

3D Gaussian Splatting (3DGS) has established a new paradigm in real-time rendering by offering an unprecedented balance between photographic quality and execution speed. However, this approach suffers from significant structural limitations: lighting is baked directly into the primitives, the implicit geometry of Gaussians cannot be exported to traditional modelling tools, and the representation of sharp surfaces requires an exponential accumulation of primitives.

This report presents *NURBS-Splatting*, a differentiable framework that substitutes discrete Gaussians with continuous NURBS surfaces, and replaces *Spherical Harmonics* with the Cook-Torrance microfacet model driven by explicit light sources. This combination aims to produce geometry that is directly usable in a CAD pipeline and compatible with dynamic relighting. To overcome the chaotic loss landscape induced by the joint optimisation of geometric parameters and PBR properties, we introduce an *Annealed Gaussian Smoothing* scheme that progressively convexifies the search space. We validate the convergence of the method on synthetic scenes with controlled initialisation and analyse the stability of the gradient flow across all optimised parameters.

INTRODUCTION

1.1 CONTEXT AND MOTIVATION

The reconstruction and representation of 3D scenes from images is a central problem in computer vision and computer graphics. The emergence of the *Neural Radiance Field* (NeRF) (Mildenhall et al. 2020) marked a paradigmatic shift by enabling the synthesis of novel views with photographic quality from a set of calibrated images, encoding the scene in the weights of an implicit neural network. Despite its remarkable results, NeRF suffers from prohibitive training and inference times that have long hindered its practical applications.

3D Gaussian Splatting (3DGS) (Kerbl et al. 2023) resolved this bottleneck by substituting the implicit neural representation with a set of explicit 3D Gaussian primitives, optimised by gradient descent and differentially rasterised. This approach achieves real-time performance while maintaining visual quality competitive with NeRF-based methods.

However, the performance of 3DGS comes at the cost of near-zero physical interpretability. Gaussians are smooth, convex functions that carry meaning only visually: they define no explicit surface, their lighting is frozen in their colour coefficients, and they cannot be directly imported into a standard CAD modelling tool. This structural opacity is a major obstacle to the integration of differentiable splatting into industrial production and digital creation pipelines.

1.2 LIMITATIONS OF EXISTING APPROACHES

Three fundamental limitations of 3DGS can be identified with respect to its use in a production pipeline:

Baked lighting. The colours of Gaussians are parameterised by *Spherical Harmonics* (SH) that encode the apparent radiance of the scene at capture time. Any modification of the lighting of the reconstructed scene requires a full re-optimisation, making dynamic relighting impossible.

Geometric inefficiency. As Gaussians are functions with unlimited support and smooth falloff, modelling sharp surfaces or hard edges requires an exponential accumulation of primitives. The number of Gaussians required grows rapidly with the geometric complexity of the scene, at the expense of representational compactness.

Lack of explicit geometry. Gaussians never define a closed surface or coherent topology. It is therefore impossible to extract a usable mesh, perform boolean operations, or import the

reconstructed scene into a modelling tool such as Blender, Maya, or an industrial CAD application.

| 1.3 CONTRIBUTIONS

This work makes the following contributions:

1. **A differentiable NURBS-based representation framework.** We substitute 3D Gaussians with continuous parametric NURBS surfaces, whose control points, knot vectors, and rational weights are optimised directly by gradient descent from photometric signals.
2. **Integration of the Cook-Torrance model.** We replace SH with the Cook-Torrance BRDF driven by explicit light sources, enabling physically correct rendering and dynamic relighting a posteriori.
3. **A coarse-to-fine optimisation scheme via Annealed Gaussian Smoothing.** To overcome local minima induced by the high frequency of PBR specularities and initial geometric misalignments, we apply a dynamically decaying low-pass filter to both the rendered and target images prior to loss computation.

RELATED WORKS

2.1 DIFFERENTIABLE RENDERING: FROM NERF TO 3DGS

Differentiable rendering consists of formulating the image formation process in a way that allows backpropagation of the photometric loss gradient towards the scene parameters. NeRF(Mildenhall et al. 2020) popularised this approach by encoding volumetric density and colour in an MLP, trained by minimising the multi-view reconstruction error via *volume rendering*. Its speed limitations motivated numerous variants, notably the use of hybrid explicit representations (voxel grids, feature planes).

3D Gaussian Splatting(Kerbl et al. 2023) represents the logical culmination of this trend towards explicit representations: the scene is modelled as a cloud of anisotropic 3D Gaussians, each endowed with a position, a covariance matrix, an opacity, and SH coefficients. These primitives are projected onto the image plane by differentiable rasterisation and composited in depth order. The algorithm incorporates an adaptive densification and pruning mechanism (*Adaptive Density Control*) that evolves the number and distribution of primitives throughout optimisation.

Recent works have sought to improve the geometric properties of 3DGS along several directions. Some have explored alternative primitives — 2D Gaussians(Huang et al. 2024), triangular primitives(Held et al. 2025a) leveraging hardware GPU rasterisers — to improve the representation of planar surfaces or reduce rasterisation cost. *3D Convex Splatting*(Held et al. 2025b) replaces Gaussians with smooth 3D convexes, offering improved capacity to represent sharp edges. The MCMC formulation(Kheradmand et al. 2025) of 3DGS reinterprets densification as a Monte Carlo process, providing a more rigorous theoretical framework for controlling primitive topology. More recently, *Geometry Grounded Gaussian Splatting*(Zhang et al. 2026) introduces a multi-view geometric consistency loss, forcing primitives to converge towards a physically plausible geometry that is coherent across views — an idea we identify as a natural extension of our approach.

2.2 NURBS MODELLING AND CAD

Non-Uniform Rational B-Splines (NURBS)(Piegl and Tiller 1997) are the industrial standard for the representation of parametric surfaces in CAD. Introduced in the 1970s and formalised by Piegl and Tiller, they provide a unified representation of curves and surfaces capable of exactly representing conic sections (spheres, cylinders, cones) without approximation. Unlike

implicit neural representations, NURBS surfaces define an explicit topology and are directly importable into any standard modelling software.

| 2.3 PHYSICAL ILLUMINATION (PBR / BRDF)

Physically Based Rendering (PBR) relies on modelling the light-matter interaction through Bidirectional Reflectance Distribution Functions (BRDFs). The Cook-Torrance microfacet model (Cook and Torrance 1982) decomposes this interaction into three terms: the microfacet normal distribution (D), the Fresnel term (F), and the geometric masking-shadowing function (G). This model can simulate a wide range of physical materials (dielectrics, metals, plastics) with intuitive roughness and metallicity parameters.

In the context of 3DGS, the work most directly related to ours is that of Gao et al., *Relightable 3D Gaussians* (Gao et al. 2024), which extends Gaussian primitives by associating them with normal vectors, BRDF parameters, and a representation of incident lighting. Their pipeline decomposes the photometric signal into material and lighting components via differentiable rendering, and introduces a point-based ray tracing method using bounding volume hierarchies (BVH) to simulate shadow effects. Our approach shares the same motivation — making the scene relightable through physical decomposition — but differs fundamentally by substituting Gaussian primitives with continuous NURBS surfaces, shifting the resolution of the problem from the domain of discrete primitive estimation to that of explicit parametric surface reconstruction.

METHOD

3.1 PIPELINE OVERVIEW

Our framework ultimately aims to take as input a set of calibrated images of a scene and produce as output a set of NURBS surfaces with their associated material properties (albedo, roughness, metallicity), directly exportable to CAD modelling tools and compatible with dynamic relighting. The target pipeline architecture is illustrated in Figure 1: scene structure extraction via SfM, light source estimation, patch topology management via MCMC, tessellation, PBR rendering, and optimisation by backpropagation.

In the current state of the work, the implemented pipeline covers the central components of this architecture: differentiable tessellation of NURBS surfaces, rendering via the Cook-Torrance model with explicit light sources, and joint optimisation of all geometric and material parameters. The upstream components — SfM reconstruction and automatic topology management via MCMC — constitute the primary axes of future work. The experiments presented in this report therefore rely on a controlled initialisation described in Section 3.2.2, which allows the convergence capacity of the optimisation core to be isolated and validated independently of these modules.

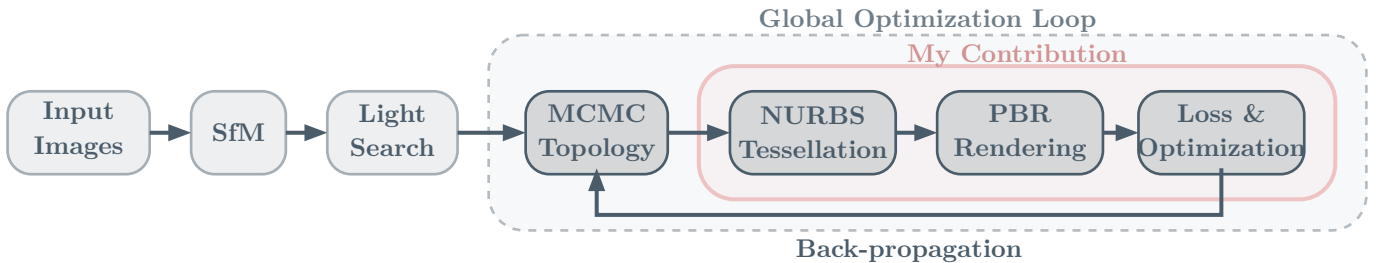


Figure 1: Architecture of the NURBS-backed PBR pipeline.

3.2 GEOMETRIC REPRESENTATION VIA NURBS

DEFINITION AND PROPERTIES

A biparametric NURBS surface is defined by:

$$S(u, v) = \frac{\sum_{i,j} N_{i(u)} N_{j(v)} w_{\{i,j\}} P_{\{i,j\}}}{\sum_{i,j} N_{i(u)} N_{j(v)} w_{\{i,j\}}}$$

where $N_{i(u)}$ and $N_{j(v)}$ are the B-spline basis functions of order p and q respectively, evaluated over the knot vectors U and V . The optimised quantities are:

Control points ($P_{\{i,j\}} \in \mathbb{R}^3$). They define the macroscopic spatial hull of the surface. Their (x, y, z) positions are optimised directly by gradient descent, constituting the primary geometric parameters of the model.

Knot vectors (U, V). They control the topology and the localisation of parametric discontinuities. Optimising the knots allows sharp edges and curvature discontinuities to be modelled, which is impossible with smooth Gaussians. They constitute an additional topological degree of freedom compared to classical splatting approaches.

Rational weights ($w_{\{i,j\}} \in \mathbb{R}^+$). The rationality of NURBS enables the exact representation of conic sections (spheres, cylinders, tori) without polynomial approximation, unlike non-rational B-splines. Optimising the weights refines the local curvature of the surface.

CONTROLLED INITIALISATION AND CONVERGENCE VALIDATION

In the current state of the work, the initialisation of NURBS is performed manually: the surfaces used to generate the synthetic target images are retrieved and perturbed by adding Gaussian noise to their control points. This controlled initialisation strategy allows the convergence capacity of the optimisation algorithm to be isolated from the question of automatic primitive placement and arrangement.

Automatic topology management — that is, determining the number of NURBS, their initial position in the scene, and their interactions — is delegated to a MCMC-based mechanism, whose integration constitutes a major axis of future work.

DIFFERENTIAL PROPERTIES OF NURBS

Unlike Gaussians, NURBS surfaces define a closed and continuous topology, guaranteeing a watertight geometry exploitable by mesh reconstruction algorithms and CAD boolean operations. The surface normal $\mathbf{n}(u, v)$, required for BRDF evaluation, is obtained analytically via the cross product of the partial derivatives:

$$\mathbf{n}(u, v) = \frac{\partial S}{\partial u} \times \frac{\partial S}{\partial v}$$

This analytical normal is more accurate than normals estimated by finite differences on implicit primitives, and its differentiability ensures clean backpropagation towards the surface parameters.

3.3 DIFFERENTIABLE RENDERING AND ILLUMINATION MODEL

LIMITATIONS OF SPHERICAL HARMONICS

In 3DGS, the colour of each Gaussian is parameterised by *Spherical Harmonics* that encode the apparent directional radiance. This approach is fundamentally limited: lighting is baked into the primitives at training time. Any modification of the scene’s lighting configuration requires a full re-optimisation. Furthermore, the use of SH prevents the model from distinguishing colour variations due to the intrinsic properties of the material from those induced by lighting: an object in shadow and the same object in light will be encoded as two distinct entities.

Our approach replaces SH with explicit light sources and the Cook-Torrance model, forcing the algorithm to decompose the photometric signal into separate contributions from geometry, material, and lighting.

COOK-TORRANCE MICROFACET MODEL

The Cook-Torrance BRDF writes:

$$f_r = k_d \frac{c}{\pi} + k_s \frac{D(\mathbf{h}) \cdot F(\mathbf{v}, \mathbf{h}) \cdot G(\mathbf{l}, \mathbf{v}, \mathbf{h})}{4(\mathbf{n} \cdot \mathbf{l})(\mathbf{n} \cdot \mathbf{v})}$$

where $\mathbf{l}$, $\mathbf{v}$, $\mathbf{h}$, and $\mathbf{n}$ denote the light, view, and half-direction vectors and the surface normal respectively. The three terms of the specular numerator are:

Normal distribution $D(\mathbf{h})$. The *GGX Normal Distribution Function* (NDF) models the statistical distribution of microfacet normals around the macroscopic normal. It is parameterised by roughness α , optimised by gradient descent:

$$D_{\text{GGX}(\mathbf{h})} = \frac{\alpha^2}{\pi((\mathbf{n} \cdot \mathbf{h})^2(\alpha^2 - 1) + 1)^2}$$

Fresnel term $F(\mathbf{v}, \mathbf{h})$. The Schlick approximation models the variation of reflectivity as a function of the angle of incidence:

$$F(\mathbf{v}, \mathbf{h}) = F_0 + (1 - F_0)(1 - \mathbf{v} \cdot \mathbf{h})^5$$

where F_0 is the reflectivity at normal incidence, parameterised by the metallicity of the material.

Geometric function $G(\mathbf{l}, \mathbf{v}, \mathbf{h})$. The Smith formulation with the GGX model simulates self-occlusion of microfacets at grazing angles:

$$G(\mathbf{l}, \mathbf{v}, \mathbf{h}) = G_{\text{GGX}(\mathbf{n} \cdot \mathbf{l})} \cdot G_{\text{GGX}(\mathbf{n} \cdot \mathbf{v})}$$

Each NURBS surface carries its own material parameters — albedo c , roughness α , and metallicity F_0 — optimised jointly with its geometric parameters.

TONE MAPPING: FROM HDR TO DISPLAY SPACE

The Cook-Torrance model operates in HDR (*High Dynamic Range*) space: the computed radiance values can greatly exceed the $[0, 1]$ interval representable on a standard display, particularly in the presence of intense specularities. A tone mapping step is therefore necessary to compress these values into LDR display space while preserving perceptual luminance relationships.

We use the filmic approximation of the ACES (*Academy Color Encoding System*) curve (Narkowicz 2015), proposed by Hill as an analytically simplified version of the full ACES pipeline (RRT + ODT). It is expressed, for each colour channel, by the rational function:

$$f_{\text{ACES}(x)} = \frac{x(ax + b)}{x(cx + d) + e}$$

with coefficients $a = 2.51$, $b = 0.03$, $c = 2.43$, $d = 0.59$, $e = 0.14$, clamped to $[0, 1]$. This polynomial approximation has several desirable properties in our context: it is differentiable over its domain, allowing gradient backpropagation through the tone mapping step; it compresses highlights smoothly and progressively rather than by hard clipping, which is essential for specular gradients to remain informative early in optimisation; and it introduces a slight contrast boost in the midtones, improving perceptual correspondence with photographic target images.

The tone mapping step is applied at the end of the pipeline, after PBR rendering and before loss computation, so that the comparison with the target image is performed in the same colour space.

| 3.4 COARSE-TO-FINE OPTIMISATION

LOSS LANDSCAPE AND THE LOCAL MINIMA PROBLEM

The joint optimisation of NURBS parameters and PBR properties constitutes a particularly challenging problem. Two main sources of non-convexity coexist:

High-frequency specularities. The specular highlights of the Cook-Torrance model are highly localised functions of geometry and material parameters. For a slightly incorrect geometric configuration, the loss gradient with respect to specularities can point in a direction entirely opposite to the desired descent direction.

Initial geometric misalignments. When control points are initialised with significant noise, the rendered surface may not overlap with the target, making the gradient signal very weak or noisy.

The combination of these two effects leads to a chaotic loss landscape scattered with deep local minima, in which a first-order optimiser such as Adam struggles to converge.

ANNEALED GAUSSIAN SMOOTHING

To convexify this landscape, we apply a Gaussian smoothing filter $\mathcal{G}_\sigma$ to both sides of the loss before its computation:

$$\mathcal{L} = \text{Huber}_\delta(\mathcal{G}_\sigma * C, \mathcal{G}_\sigma * C^*)$$

where C is the rendered image, C^* the target image, and Huber_δ the Huber loss with threshold δ .

The parameter σ controls the filter width and decays throughout optimisation according to a *coarse-to-fine* schedule:

$$\sigma(t) = \max\left(0.1, 3.0 \cdot \left(1.0 - \frac{t}{t_{\text{end}}}\right)\right)$$

where t is the current iteration and t_{end} the iteration at which the blur becomes inactive.

Coarse-to-fine dynamics. For a large σ , both images — rendered and target — are heavily blurred. High frequencies (specularities, edges) vanish, and the gradient signal is dominated by the low-frequency structure of the scene (global object placement, general light distribution). The optimiser can thus secure the global topological placement of NURBS patches. As σ decays towards zero, the high-frequency details of the BRDF are progressively reintroduced, allowing the optimiser to refine geometry and local material properties.

ROBUST HUBER LOSS

The Huber loss (Huber 1964) is preferred over MSE for its robustness to outliers:

$$\text{Huber}_\delta(x) = \begin{cases} \frac{1}{2}x^2 & \text{if } |x| \leq \delta \\ \delta(|x| - \frac{1}{2}\delta) & \text{otherwise} \end{cases}$$

In our context, outliers correspond primarily to specular pixels that are poorly fitted early in optimisation. The quadratic-to-linear transition of the Huber loss limits the influence of these pixels on the global gradient, preventing specularities from saturating the backpropagation signal at the expense of geometric placement.

EXPERIMENTS

4.1 IMPLEMENTATION DETAILS

The pipeline is implemented in Python with PyTorch. Differentiable tessellation of NURBS surfaces and analytical normal computation rely on the Kaolin library. Differentiable rasterisation is handled by nvdiffrast. Optimisation is performed with Adam over 2500 iterations, with distinct learning rates per parameter group, reflecting their respective scales and sensitivities:

Parameter	Learning rate
Control points	10^{-3}
Rational weights	10^{-6}
Knots U	5×10^{-3}
Knots V	5×10^{-3}
Albedo	10^{-3}

4.2 PROTOCOL: SYNTHETIC SCENES AND CONTROLLED INITIALISATION

In the absence of an automatic primitive placement mechanism, we adopt a validation protocol on controlled synthetic scenes. Target images are generated by a reference renderer from a known NURBS scene. The algorithm receives as input the perturbed scene — that is, the same NURBS patches whose control points, albedo, roughness, and metallicity have been corrupted by Gaussian noise — and must recover the original configuration from the multi-view photometric signal alone.

This protocol isolates and validates the optimisation component of the pipeline, independently of the problem of automatic primitive initialisation and placement.

Experiments are conducted on a scene composed of three spheres of different colours. Four views of the scene are used as supervision.

4.3 QUALITATIVE RESULTS: MULTI-VIEW CONVERGENCE

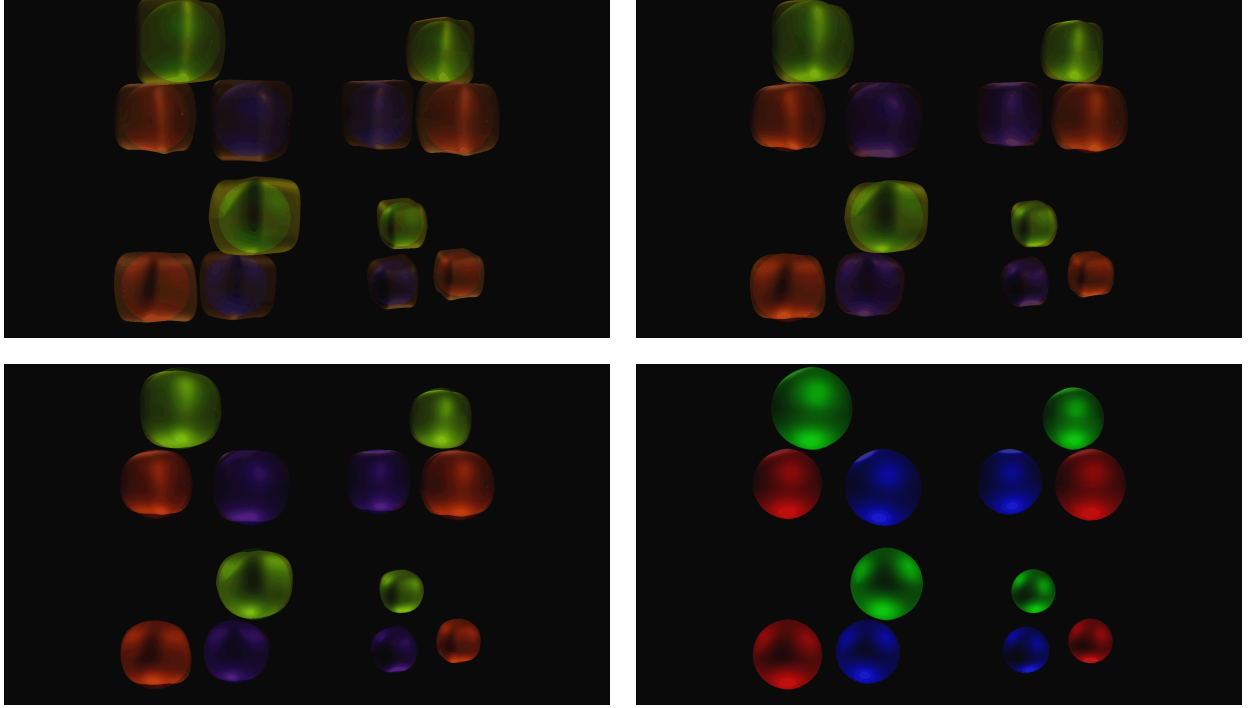


Figure 2: Simultaneous coarse-to-fine optimisation across four views.

Figure 2 illustrates the convergence of the method across four simultaneous views. Starting from a noisy initialisation, the pipeline progressively recovers the target geometry, as evidenced by the increasing visual quality of the renderings over iterations. All four views converge simultaneously, validating the multi-view consistency of the optimisation.

We observe that convergence is first driven by the low-frequency structure of the scene — global object placement, light distribution — before surface details and specular effects are progressively resolved. This behaviour is consistent with the expected dynamics of the Annealed Smoothing scheme.

4.4 GRADIENT FLOW ANALYSIS AND STABILITY

Gradient Norm (log10)

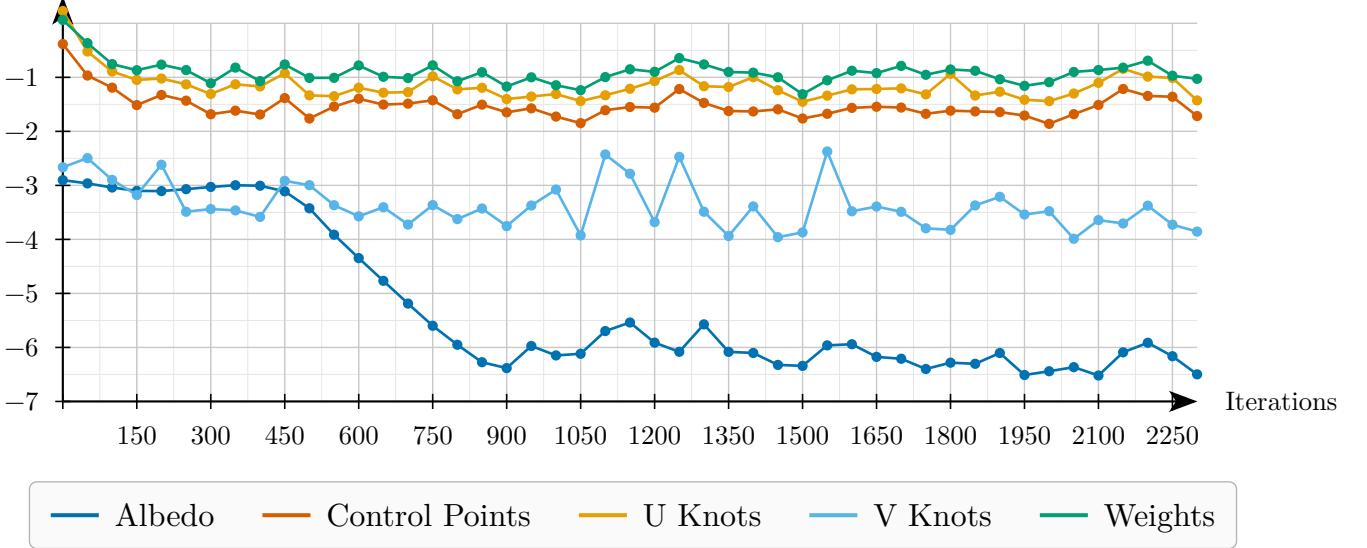


Figure 3: Evolution of the logarithmic gradient norm during optimization.

Figure 3 shows the evolution of the gradient norm (in log scale) for each parameter group over 2250 iterations. Several observations are noteworthy.

Global stability. The absence of catastrophic spikes throughout the optimisation trajectory confirms that the Annealed smoothing filter effectively regularises the loss landscape. Gradients decay in a monotone and regular fashion, a sign of stable convergence.

Parameter hierarchy. Control points exhibit the highest gradients early in optimisation, consistent with the fact that the initial geometric error (initialisation noise) dominates the loss signal. Albedo parameters converge faster than geometric parameters, suggesting that the model first resolves coarse photometric properties before refining geometry.

Convergence of knots and weights. The gradients of knot vectors (Knots U, Knots V) and rational weights remain lower but stable throughout optimisation. This indicates that these fine topological parameters are only marginally modified, which is expected for a relatively simple synthetic scene.

4.5 ABLATION: IMPACT OF ANNEALED SMOOTHING

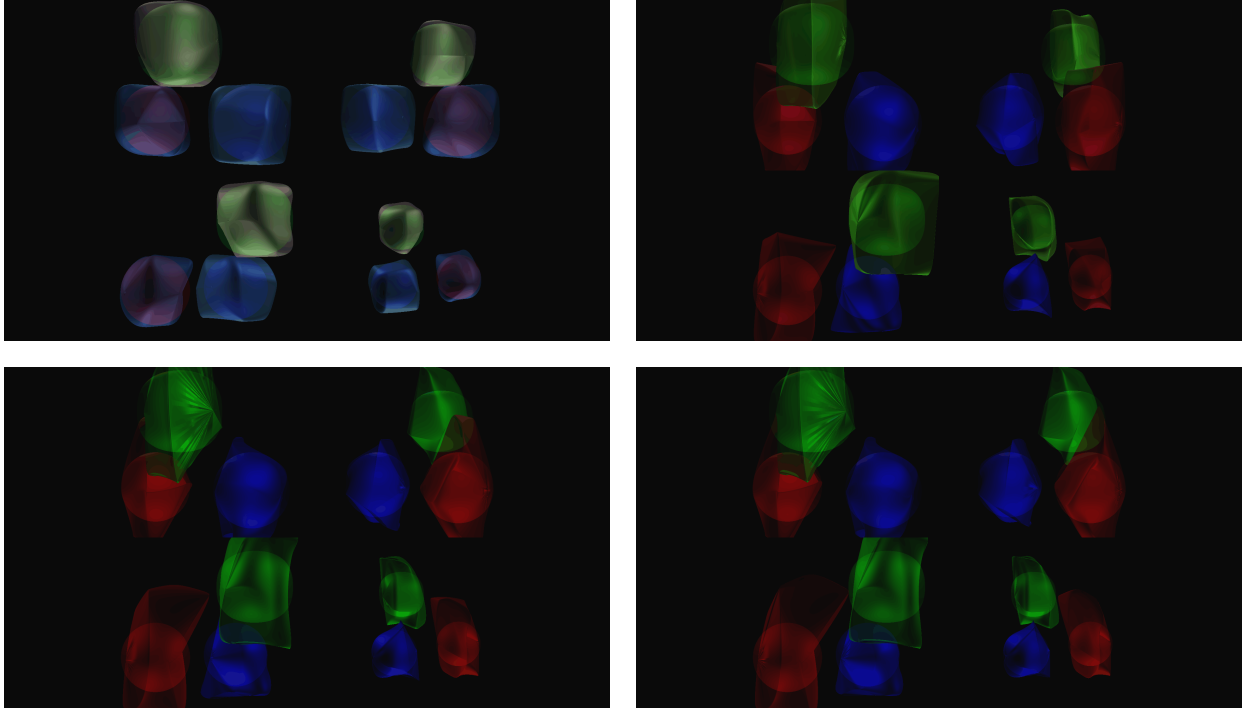


Figure 4: Optimisation with Huber loss only, across four views.

Without the Annealed Gaussian Smoothing scheme, optimisation driven by the Huber loss alone fails to converge geometrically: the NURBS surfaces do not reach the target configuration and progressively degrade over iterations. Only coarse photometric properties — notably the dominant surface colour — are partially recovered by the model. This result confirms that the high-frequency specularities of the Cook-Torrance model saturate the geometric gradient signal from the very first iterations, trapping the optimiser in a local minimum. Annealed Gaussian Smoothing is therefore an indispensable component for geometric convergence, and not merely an optional refinement.

DISCUSSION AND LIMITATIONS

| 5.1 PROOF OF CONCEPT VALIDATION

The obtained results validate the core hypothesis of this work: NURBS surfaces can be optimized directly from a multi-view differentiable photometric signal. The method converges stably on the synthetic validation protocol, and the gradient tracking confirms that the Annealed Smoothing scheme effectively fulfills its role as a loss landscape regularizer.

The integration of the Cook-Torrance model with explicit light sources is fully functional: the pipeline correctly decomposes the photometric signal into geometric and lighting contributions, paving the way for a posteriori dynamic relighting without requiring re-optimization.

| 5.2 CURRENT LIMITATIONS

Manual Initialization. The current experimental protocol relies on a controlled initialization where the NURBS patches are derived from the target scene. Automating this initialization—determining the number of patches, their positions, and their initial orientations directly from images—is a fundamental prerequisite for applying the method to real-world scenes.

Simple Validation Scenes. The experiments are conducted on geometrically simple synthetic scenes. Generalization to complex scenes with multiple occlusions, diverse materials, and non-trivial topologies remains to be demonstrated.

Lack of Topological Adaptation Mechanism. The number and resolution of the NURBS patches are fixed prior to optimization. A dynamic adaptation mechanism—subdividing poorly fitted patches and merging redundant ones—is necessary to allow the representation to evolve according to the local geometric complexity of the scene.

Limited Quantitative Validation. In the absence of testing on standard benchmarks (such as NeRF Synthetic or Tanks and Temples), it is not currently possible to quantitatively compare the performance of NURBS-Splatting against state-of-the-art methods.

| 5.3 FUTURE WORK

Multi-View Geometric Consistency Loss. Recent works in geometry-grounded Gaussian Splatting have demonstrated the benefit of a multi-view geometric consistency loss to constrain the reconstruction toward physically plausible geometry. Adapting this concept to the NURBS framework—leveraging the available analytical normals and the explicit surface topology—constitutes a natural and promising extension.

Ray Tracing as the Rendering Engine. The current differentiable rasterization is an approximation of actual light transport: it does not model cast shadows, inter-object reflections, or caustics. Replacing the rasterizer with a differentiable ray tracing engine would unlock the full potential of the Cook-Torrance model, moving toward physically exact rendering that aligns with the method’s goal of physical interpretability.

Dynamic Topology via MCMC. Integrating a Markov Chain Monte Carlo mechanism to govern the translation, creation, and deletion of NURBS patches would automate initialization and topological adaptation. This mechanism would replace the current manual initialization and stands as the highest priority for future work.

Adaptive Parametric Resolution. Building upon the MCMC engine, it would be possible to dynamically adjust the control point density of each patch based on the local geometric complexity of the scene, ensuring a compact and highly efficient representation.

CONCLUSION

This report presents NURBS-Splatting, a differentiable representation framework that substitutes the discrete Gaussians of 3DGS with continuous parametric NURBS surfaces, coupling their optimization with a physically based illumination model grounded in the Cook-Torrance BRDF.

The core contribution of this work is twofold. First, we demonstrate the feasibility of optimizing NURBS surfaces via a multi-view differentiable photometric signal. Second, we introduce an *Annealed Gaussian Smoothing* scheme that stabilizes the joint geometry-PBR optimization by progressively convexifying the loss landscape.

These results establish a proof of concept for an approach that, once equipped with an automatic initialization and topological adaptation mechanism, paves the way for a 3D scene representation that is simultaneously photometrically accurate, physically interpretable, and directly usable within traditional CAD and digital modeling pipelines.

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